

Unlike the nucleus of a eukaryotic cell, this dense region of DNA in a bacterium, called the nucleoid, is not bounded by membrane (see Figure 6.5).

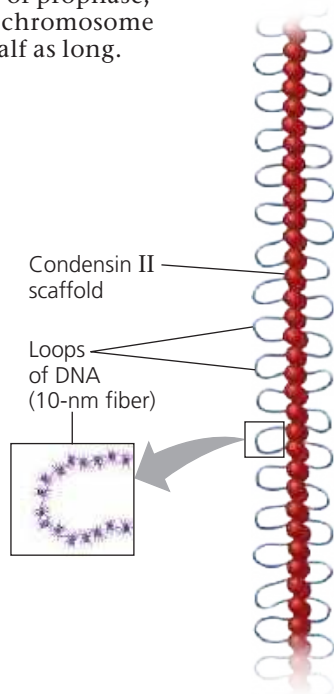
Each eukaryotic chromosome contains a single linear DNA double helix that, in humans, averages about  $1.5 \times 10^8$  nucleotide pairs. This is an enormous amount of DNA relative to a chromosome's condensed length. If completely stretched out, such a DNA molecule would be about 4 cm long, thousands of times the diameter of a cell nucleus—and that's not even considering the DNA of the other 45 human chromosomes!

In the cell, eukaryotic DNA is precisely combined with a large amount of protein. The exact way in which this complex of DNA and protein, called **chromatin**, fits into the nucleus has long been debated. **Figure 16.23** outlines the current view of the organization of chromatin during interphase and how it is condensed during mitosis into the metaphase chromosome. Study this figure carefully before reading further.

Chromatin undergoes striking changes in how densely it is packed during the course of the cell cycle (see Figure 12.7). In interphase cells stained for light microscopy, the

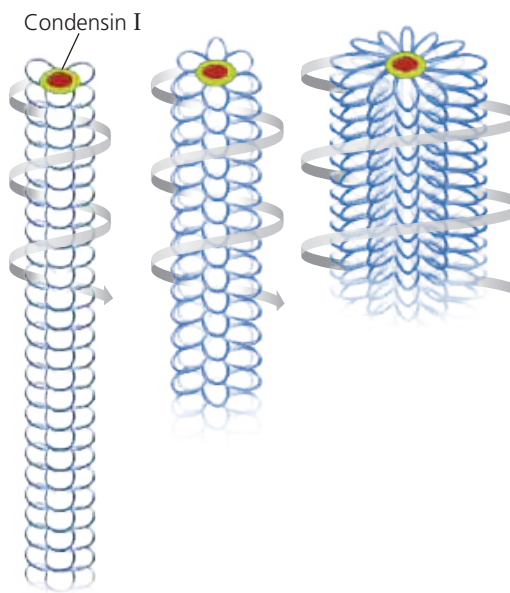
## Prophase

When mitosis begins, DNA replication has already occurred, so each chromosome consists of two sister chromatids. During prophase, the chromatin of each sister chromatid begins to condense. Two related proteins called condensin II and condensin I play important roles. First, condensin II proteins (red) bind to the 10-nm fiber of DNA and form DNA loops that get larger and larger. The condensin II proteins form a central scaffold from which the loops extend. As the loops grow, the chromosome gets wider and shorter. By the end of prophase, the chromosome is half as long.



## Prometaphase

When prometaphase begins, condensin I proteins (green) bind to DNA outside the central scaffold, making smaller loops (not shown) out of the larger loops generated by condensin II. The process continues, with more and more loops extending outward, and the chromosome getting denser, shorter, and wider. The scaffold itself also begins to twist into a helix (suggested by the curved gray arrows), allowing even more loops per given length of chromosome.



## Metaphase

At metaphase, the chromosome is at its most dense, with the most loops per turn, and therefore is at its shortest length. The two sister chromatids are fully condensed.

